

1 Conclusion and Future Work

1.a Summary of Thesis Findings

Recap of the feasibility and performance demonstrated by the SDR-based OAC network.

1.b Future Research Directions

Exploring more complex OAC architectures, such as multi-antenna systems Multiple-Input Multiple-Output (MIMO) or using Reconfigurable Intelligent Surfaces (RIS)[1], [2]. Further investigation into standardized OAC protocols for compatibility with modern wireless networks (e.g., LTE/5G protocols)[3], [4].

References

- [1] N. G. Evgenidis et al., “Waveform design for over-the-air computing,” *IEEE Transactions on Wireless Communications*, pp. 1–1, 2025, ISSN: 1536-1276, 1558-2248. DOI: 10.1109/TWC.2025.3613838 arXiv: 2405.20877[cs]. Accessed: Dec. 4, 2025. [Online]. Available: <http://arxiv.org/abs/2405.20877>
- [2] J. Zhu, Y. Shi, Y. Zhou, C. Jiang, W. Chen, and K. B. Letaief, “Over-the-air federated learning and optimization,” *IEEE Internet of Things Journal*, vol. 11, no. 10, pp. 16 996–17 020, May 15, 2024, ISSN: 2327-4662, 2372-2541. DOI: 10.1109/JIOT.2024.3352280 Accessed: Dec. 4, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/10388035/>
- [3] G. Zhu, Y. Du, D. Gunduz, and K. Huang, “One-bit over-the-air aggregation for communication-efficient federated edge learning: Design and convergence analysis,” *IEEE Transactions on Wireless Communications*, vol. 20, no. 3, pp. 2120–2135, Mar. 2021, ISSN: 1536-1276, 1558-2248. DOI: 10.1109/TWC.2020.3039309 Accessed: Dec. 4, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/9272666/>
- [4] A. Şahin and R. Yang, “A survey on over-the-air computation,” *IEEE Communications Surveys & Tutorials*, vol. 25, no. 3, pp. 1877–1908, 2023, ISSN: 1553-877X, 2373-745X. DOI: 10.1109/COMST.2023.3264649 Accessed: Dec. 4, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/10092857/>